LEVEL 1

Industrial Temperature Monitoring & Safety Controller

1. PROJECT OVERVIEW

An industrial process (a furnace zone, a motor housing, a control cabinet) needs its temperature watched continuously, with a proportional cooling response and a hard alarm if the temperature becomes unsafe. This project builds that controller: an analog temperature sensor feeds the PIC16F877A's ADC, the reading is shown on an LCD, a PWM-driven cooling fan speeds up as temperature rises, and a buzzer/LED trips at a critical threshold.

2. INDUSTRIAL APPLICATION

The same acquire–display–proportionally-cool–alarm pattern is used in electrical panel cooling, motor-housing thermal protection, and process-zone temperature regulation across manufacturing plants. This handbook implements it as a low-voltage educational simulation, not a certified industrial safety device.

3. OBJECTIVES

- Configure the PIC16F877A ADC to sample an analog temperature sensor.
- Display live temperature on a 16x2 LCD.
- Generate a PWM signal whose duty cycle scales with temperature to drive a cooling fan.
- Implement a fixed critical-threshold alarm using a buzzer and LED.
- Structure the firmware into reusable ADC, LCD, and PWM functions.

4. KEY FEATURES

- Live LCD temperature readout
- Proportional PWM fan-speed control (speed rises with temperature)
- Fixed critical-threshold buzzer/LED alarm
- Modular ADC/LCD/PWM driver functions
- Stable polling loop with debounced display refresh

5. HARDWARE REQUIREMENTS

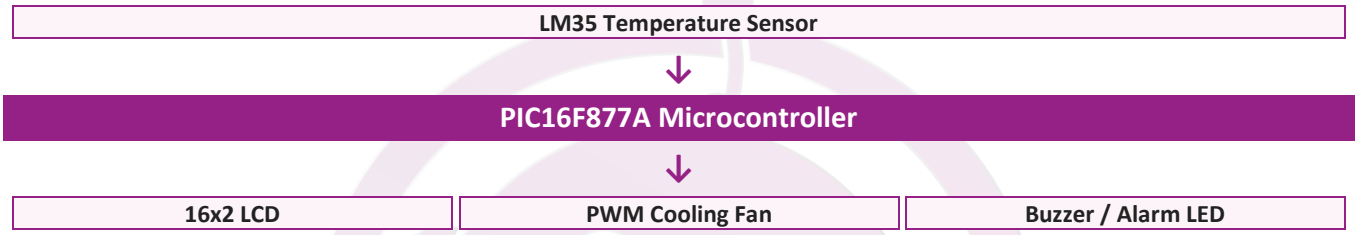
- PIC16F877A microcontroller
- LM35 analog temperature sensor
- 16x2 LCD (HD44780-compatible)
- Small 12V DC cooling fan (PWM driven via a transistor stage)
- Buzzer and alarm LED
- 20 MHz crystal oscillator with 22pF capacitors

- 5V regulated supply

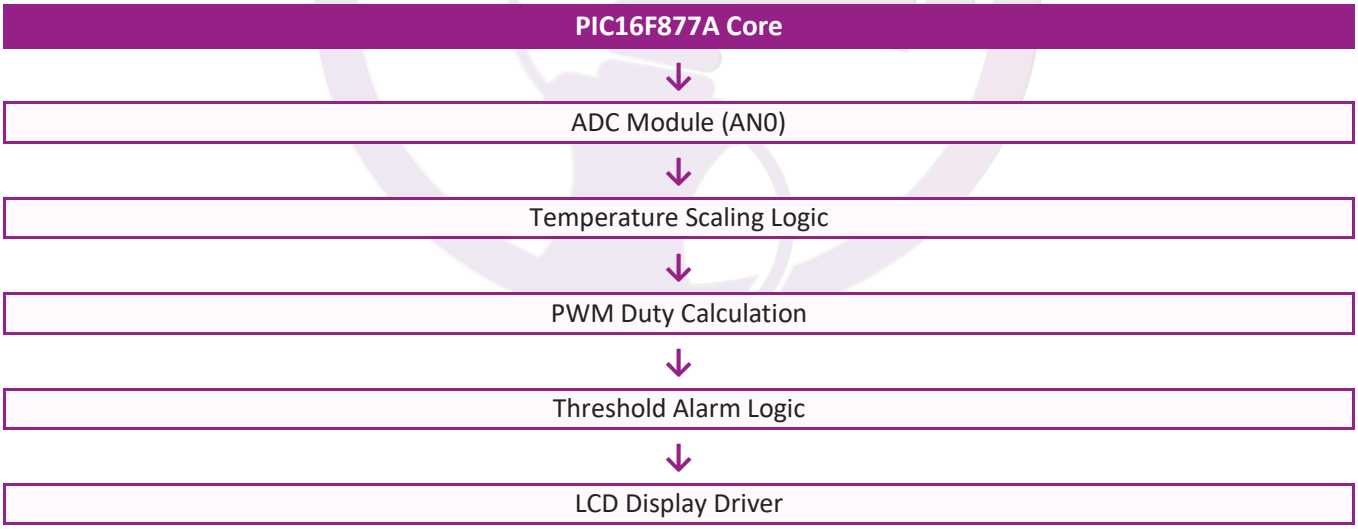
6. SOFTWARE REQUIREMENTS

- MPLAB X IDE
- XC8 Compiler
- Proteus Design Suite (for simulation)
- Embedded C (ANSI C for 8-bit PIC architecture)

7. BLOCK DIAGRAM



8. SYSTEM ARCHITECTURE



9. WORKING PRINCIPLE

The LM35 produces 10 mV per degree Celsius on AN0. The firmware triggers an ADC conversion, reads ADRESH:ADRESL, and converts the raw value to a temperature reading. This value is displayed on the LCD and also mapped linearly to a PWM duty cycle: low temperature gives a low or zero fan duty, and duty rises smoothly toward 100% as temperature approaches the critical threshold. If temperature exceeds the critical threshold, the firmware forces the fan to full duty and simultaneously activates the buzzer and alarm LED, independent of the proportional control calculation.

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10. PIC16F877A PERIPHERALS USED

Peripheral	Purpose in This Project
ADC	Samples the LM35 output on channel AN0.
Timer2 / CCP1 (PWM)	Generates the variable-duty-cycle signal driving the cooling fan.
GPIO	Drives the LCD, buzzer, and alarm LED.

11. IMPORTANT REGISTERS

Register	Purpose / Configuration
TRISA / TRISC / TRISD	Configure AN0 as analog input; LCD, buzzer, and fan-driver pins as outputs.
ADCON0 / ADCON1	Select channel AN0, set the reference, and control conversion timing.
ADRESH / ADRESL	Hold the completed 10-bit temperature reading.
T2CON / PR2	Set the PWM period for the fan drive signal.
CCP1CON / CCPR1L	Set the PWM duty cycle proportional to measured temperature.

12. PROGRAM FLOW

Initialization: GPIO, LCD, ADC, PWM

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Input Acquisition: Read Temperature via ADC

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Processing: Scale ADC Value to °C and to PWM Duty

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Decision: Temperature ≥ Critical Threshold?

↓

Output: Update LCD and Apply PWM Duty to Fan

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Fault Handling: Force Full Fan Duty + Buzzer/LED on Critical Alarm

↓

Delay and Repeat

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13. IMPLEMENTATION

Implementation Area	What to Do
Hardware Connections	Wire LM35 to AN0, LCD to PORTD in 4-bit mode, and the fan-driver transistor base to the CCP1 (RC2) pin.
Peripheral Configuration	Configure ADCON0/ADCON1 for AN0, and T2CON/PR2/CCP1CON for a ~5 kHz PWM signal.
Sensor Implementation	Write an ADC_Read() function and a temperature-scaling function based on the LM35's 10 mV/°C output.
Control Logic	Implement linear mapping from temperature to PWM duty within the normal operating band.
Output Implementation	Update the LCD every cycle; drive the fan via PWM_SetDuty().
Fault Handling	On critical threshold breach, override the calculated duty with 100% and latch the buzzer/LED.
Testing	Vary the simulated LM35 voltage in Proteus across the full operating and critical range and confirm fan and alarm response.

14. EMBEDDED C IMPLEMENTATION

PWM Initialization (Timer2/CCP1)

```
void PWM_Init(void)
{
    TRISC2 = 0;
    PR2 = 249;          /* ~5 kHz PWM at Fosc = 20 MHz, prescale 1:4 */
    T2CON = 0x05;        /* Timer2 ON, prescaler 1:4 */
    CCP1CON = 0x0C;       /* CCP1 module in PWM mode */
    CCPR1L = 0;
}
void PWM_SetDuty(unsigned int duty10bit)
{
    CCPR1L = duty10bit >> 2;
    CCP1CON = (CCP1CON & 0xCF) | ((duty10bit & 0x03) << 4);
}
```

Proportional Fan Control with Critical Override

```
void Update_Cooling(float temp_c)
{
    if (temp_c >= CRITICAL_TEMP)
    {
        PWM_SetDuty(1023);     /* force full fan speed */
        BUZZER = 1;
        ALARM_LED = 1;
        return;
    }
    BUZZER = 0;
    ALARM_LED = 0;
    unsigned int duty = (unsigned int)((temp_c / CRITICAL_TEMP) * 1023);
    PWM_SetDuty(duty);
}
```


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15. PROTEUS SIMULATION

Component List

- PIC16F877A
- LM35 (or potentiometer as a stand-in analog source)
- 16x2 LCD (LM016L)
- NPN transistor + small DC motor/fan model
- Buzzer, alarm LED
- 20 MHz crystal

Important PIC Pins

Pin	Connection
AN0 (RA0)	LM35 signal input
RC2 / CCP1	PWM output to fan driver transistor
PORTD	LCD data/control
RB0, RB1	Buzzer, alarm LED

Simulation Procedure

Build the schematic, load the compiled .hex, and run the simulation. Sweep the LM35 (or substitute potentiometer) through its full range and observe the LCD reading, the fan's simulated speed change, and the alarm activating past the critical point.

Expected Behavior

Fan duty cycle rises smoothly with temperature within the normal band, and jumps to full speed with an audible/visible alarm once the critical threshold is crossed.

Inputs to Vary During Testing

- LM35 equivalent input swept from 0°C to above the critical threshold
- Rapid step changes near the critical threshold to confirm alarm latching behavior

16. HARDWARE IMPLEMENTATION

Notes for transferring this project from Proteus simulation to a physical PIC16F877A development board:

- Use a regulated 5V supply for the PIC and LCD; the fan motor may need a separate 12V rail sharing a common ground with the PIC.

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- Drive the fan through a flyback-diode-protected transistor or small MOSFET stage — never connect a motor/fan directly to a PIC I/O pin.
- Keep the LM35 wiring short and away from the fan motor leads to reduce electrical noise on the analog input.
- Verify the crystal and capacitor values match the Proteus design before powering the physical board.
- Confirm all grounds (PIC, fan driver stage, LCD, sensor) are tied to one common ground point.

17. TEST CASES

Test	Input / Condition	Expected Result
Normal operation	Temperature well within safe range	LCD shows correct value, fan duty proportionally low, alarm off
Rising temperature	Temperature increases gradually	Fan duty increases smoothly with temperature
Near critical threshold	Temperature just below critical limit	Fan at high duty, alarm still inactive
Critical threshold reached	Temperature ≥ critical limit	Fan forced to full duty, buzzer and LED activate
Temperature falls after alarm	Temperature drops below critical limit	Alarm clears, proportional control resumes
Sensor at 0°C equivalent	LM35 input = 0V	LCD shows 0°C, fan at minimum duty, no crash
Sensor near full-scale input	LM35 input near 5V equivalent	LCD shows a high value, alarm active
Power-on state	System reset	LCD shows startup message before first valid reading

18. FAULT HANDLING

- Critical over-temperature: fan forced to full duty and buzzer/LED latched until temperature returns below threshold.
- Sensor disconnect/out-of-range reading: a raw ADC value pinned at 0 or full-scale for multiple consecutive samples is treated as a probable sensor fault and displayed as such on the LCD rather than a plausible temperature.
- LCD not responding at startup: the initialization routine includes the LM44780 power-on delay sequence to avoid sending commands before the display is ready.

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19. EXPECTED OUTPUT

Output	Expected Behavior / Display
LCD	"Temp: 42C Fan:65%" during normal proportional operation; "**ALARM**" on critical breach.
Buzzer / LED	OFF during normal and warming operation; ON continuously once the critical threshold is crossed.
PWM Fan Output	Duty cycle rises smoothly from near 0% to 100% across the normal operating band, then pinned at 100% during alarm.

20. SKILLS DEMONSTRATED

ADC Interfacing | LCD Interfacing | PWM Generation (Timer2/CCP1) | Proportional Control Logic | Embedded C | Modular Firmware Design

21. CAREER & RESUME RELEVANCE

Professional resume bullet examples describing the technical work completed in this project:

- Developed a PIC16F877A-based temperature monitoring controller combining ADC sensing, LCD readout, and PWM-driven proportional cooling.
- Implemented threshold-based alarm logic with buzzer/LED indication layered on top of proportional fan-speed control.
- Built and validated the complete sensing-to-actuation pipeline in Proteus prior to physical hardware testing.

22. INTERVIEW QUESTIONS

Question	Concise Answer
How does the PIC16F877A's ADC represent an analog voltage digitally?	It performs successive-approximation conversion producing a 10-bit result (0–1023) split across ADRESH and ADRESL, proportional to the input voltage relative to the reference.
Why does the fan speed change smoothly instead of switching on/off?	The firmware maps the measured temperature linearly onto the PWM duty cycle, so the fan response is proportional rather than binary, giving smoother thermal regulation.
What do T2CON and PR2 control in this project?	T2CON enables Timer2 and sets its prescaler; PR2 sets the period the timer counts to, and together with the prescaler they define the overall PWM frequency.
Why is the critical alarm logic separate from the proportional calculation?	So that a wiring or logic fault in the proportional path cannot suppress the safety alarm — the critical check overrides the calculated duty unconditionally.
What is the purpose of ADCON1 in this design?	It configures which PORTA pins are treated as analog versus digital and sets the voltage reference source used for conversion.
How would you detect a disconnected temperature sensor in firmware?	By checking for ADC readings pinned at the rail values (near 0 or near 1023) for several consecutive samples, which is atypical of a normally varying analog signal.

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Question	Concise Answer
Why use a transistor stage instead of driving the fan directly from a PIC pin?	PIC I/O pins cannot safely source the current a DC fan motor requires, and a transistor/MOSFET stage with a flyback diode protects the pin from back-EMF.
How would you calculate the PWM frequency for a chosen PR2 value?	PWM frequency = Fosc / [4 × (PR2+1) × TMR2 prescaler]; with Fosc = 20 MHz and prescaler 4, PR2 = 249 gives approximately 5 kHz.

23. POSSIBLE ENHANCEMENTS

The following are potential future enhancements and are not implemented in this handbook:

- Add EEPROM-based storage of the configured critical threshold.
- Add UART reporting of temperature and fan duty to a PC.
- Add hysteresis around the critical threshold to prevent alarm chattering.
- Add a second sensor channel for redundancy.

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